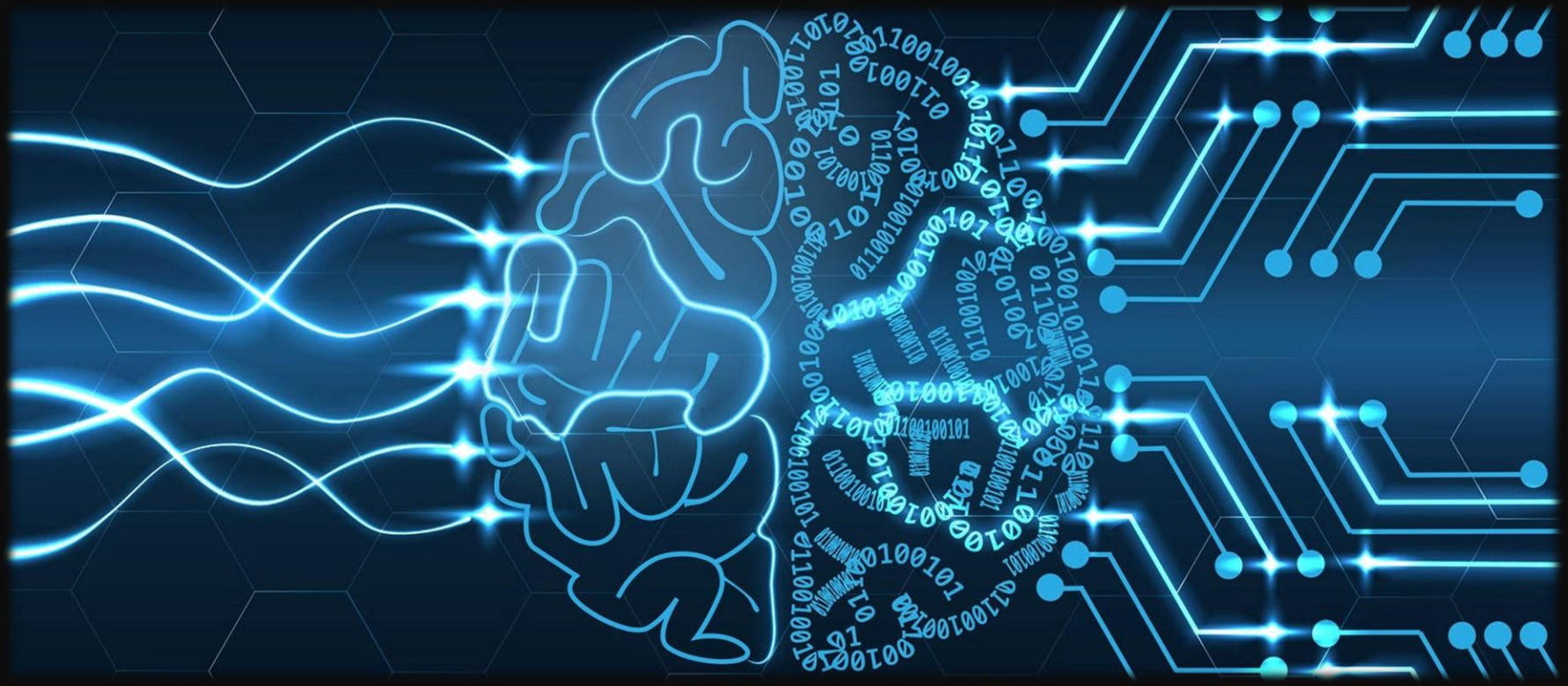
CNN Temelleri



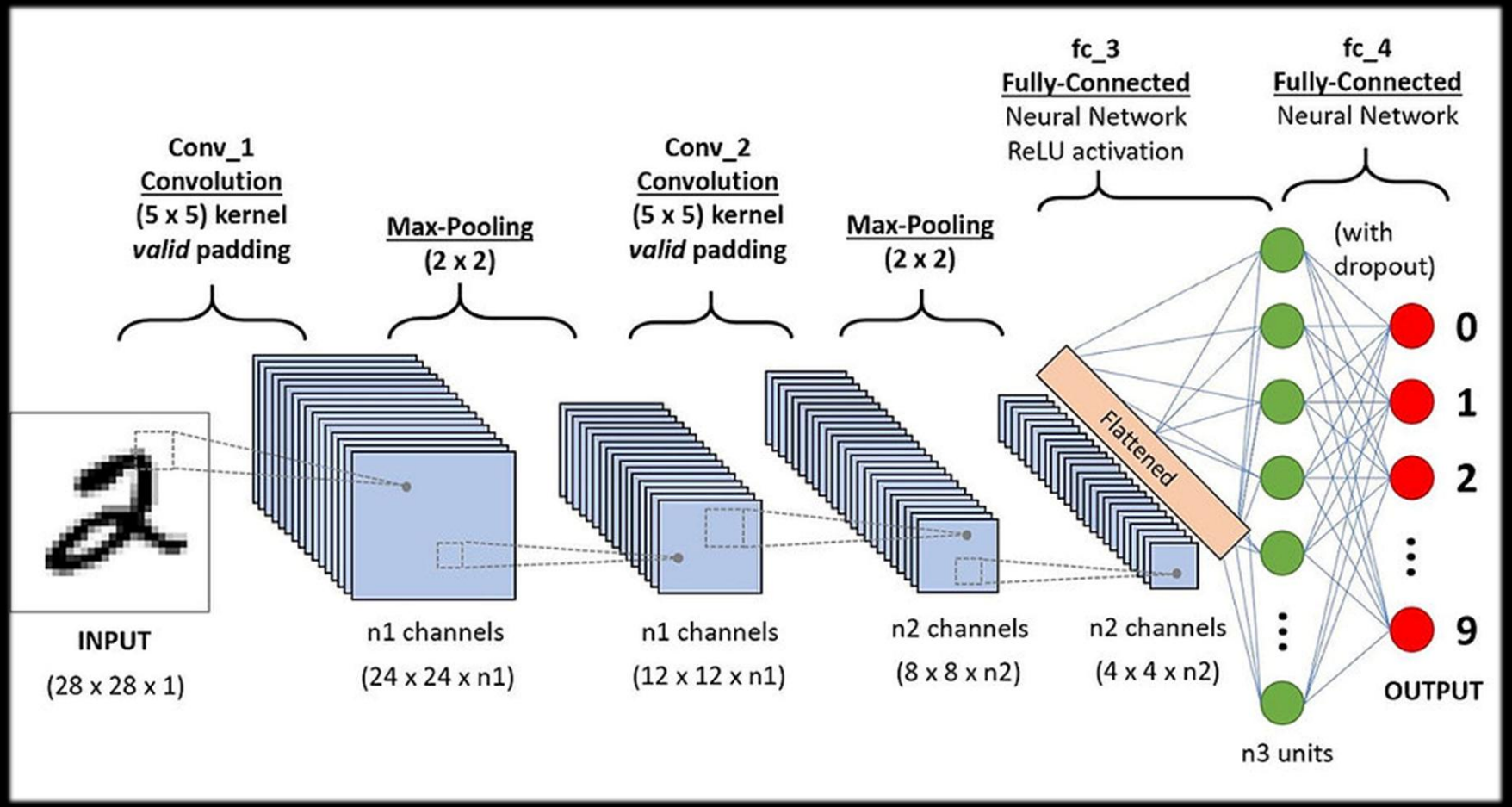
MUSTAFA YASİR KARADAĞ
2322705041

Neden CNN?

Mekansal Bilgi Kaybı (Spatial Information)

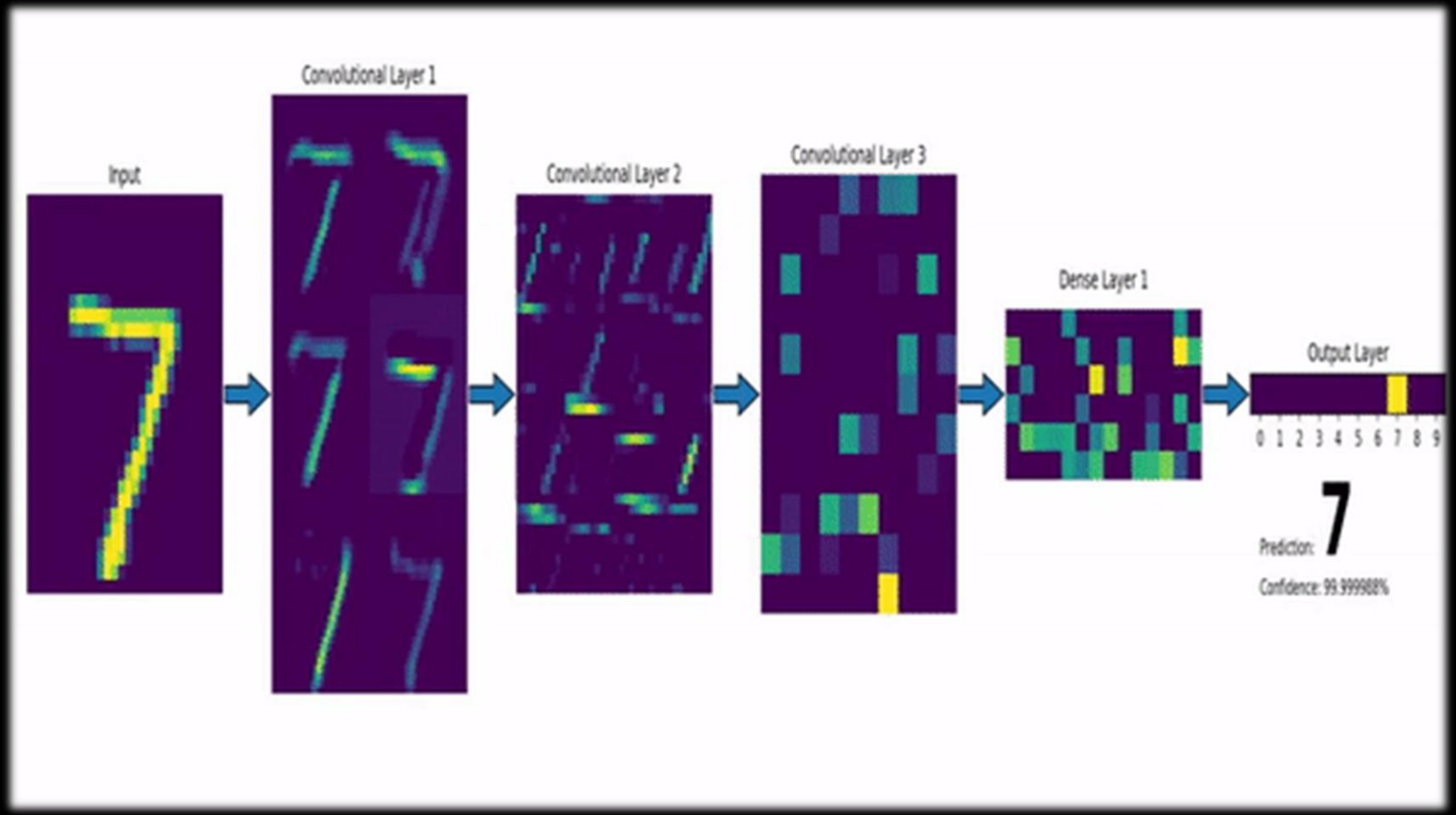
ANN: Vektörleştirme

CNN: 2B Özellik Koruma



Konvolüsyon Mantığı

Filtreler (Kernels): Özellik Yakalayıcılar
Özellik Haritası (Feature Map)
Analiz: Kenar, Köşe ve Doku Analizi

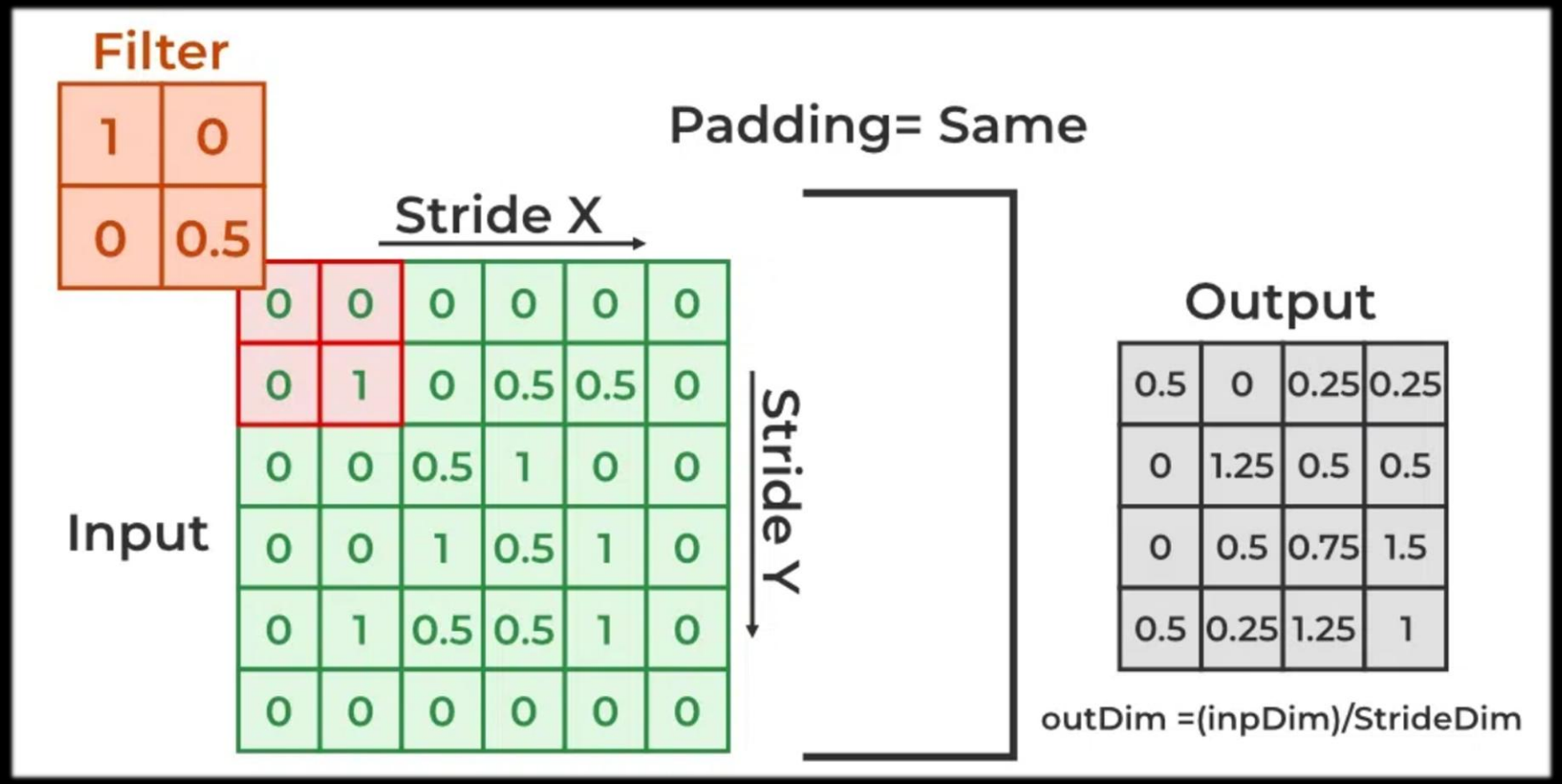


Padding ve Stride

Padding: Görsel boyutunu koruma ve kenar bilgilerini kaybetmeme

Stride: Filtrenin kaydırma adımı

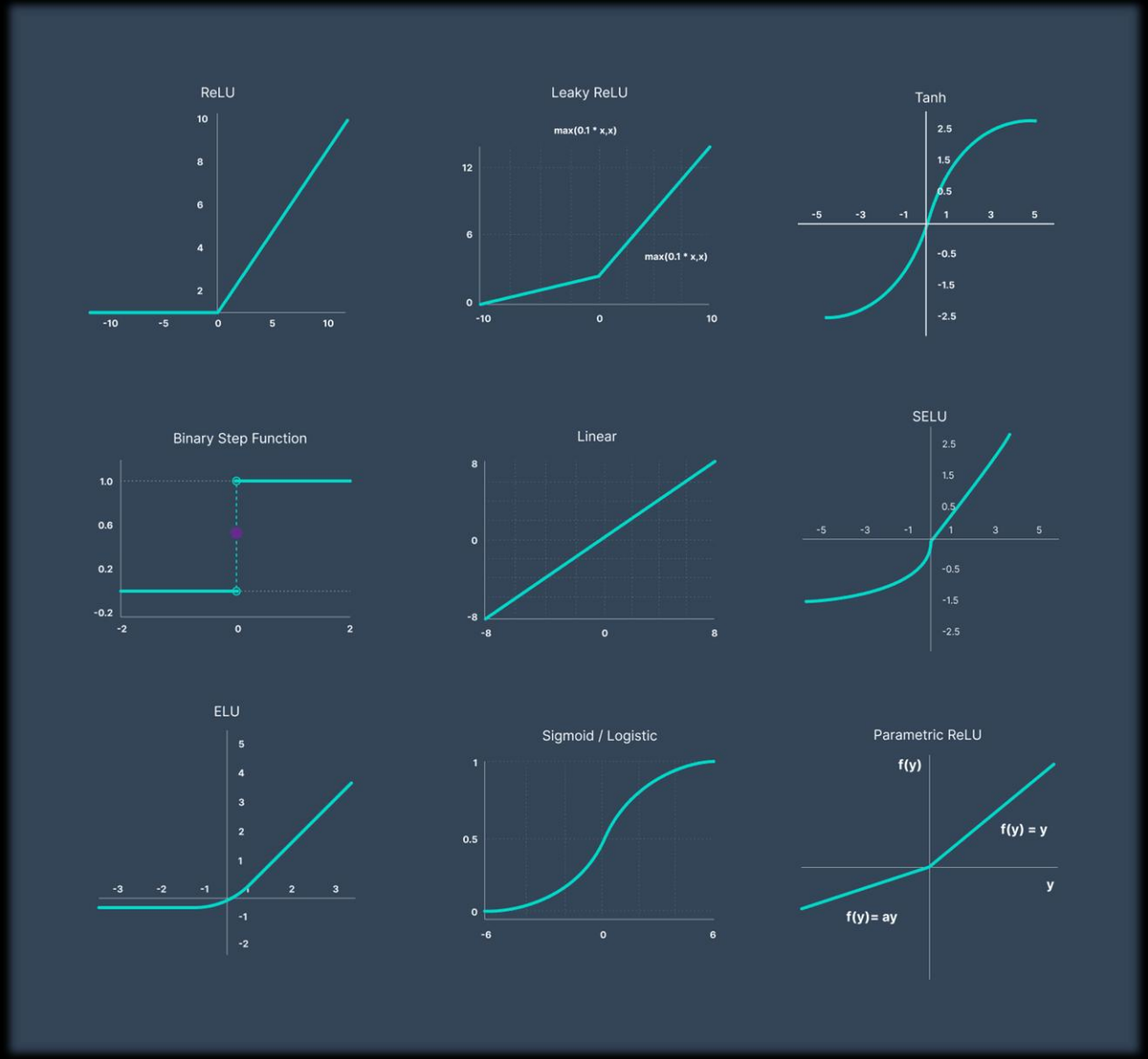
Çıkış Boyutu: $O = \frac{W-K+2P}{S} + 1$



Aktivasyon Katmanı (ReLU)

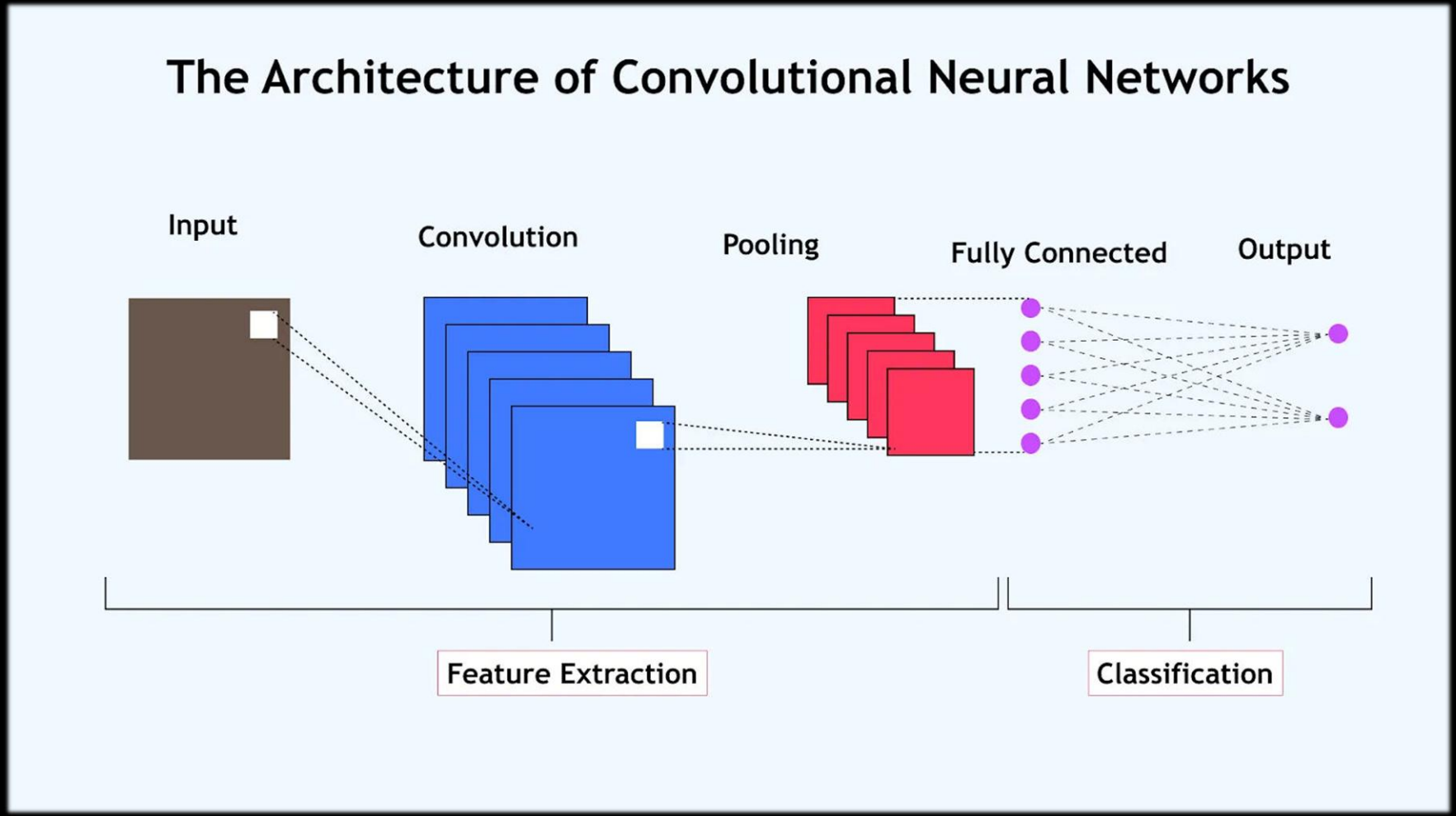
Doğrusal Olmayan Yapı: Karmaşık verileri öğrenmeyi sağlar.
Negatifleri Sıfırlama: Sıfırdan küçük değerleri eler.
Hızlı Hesaplama: Basit matematiksel yapıdadır.

$$f(x) = \max(0, x)$$



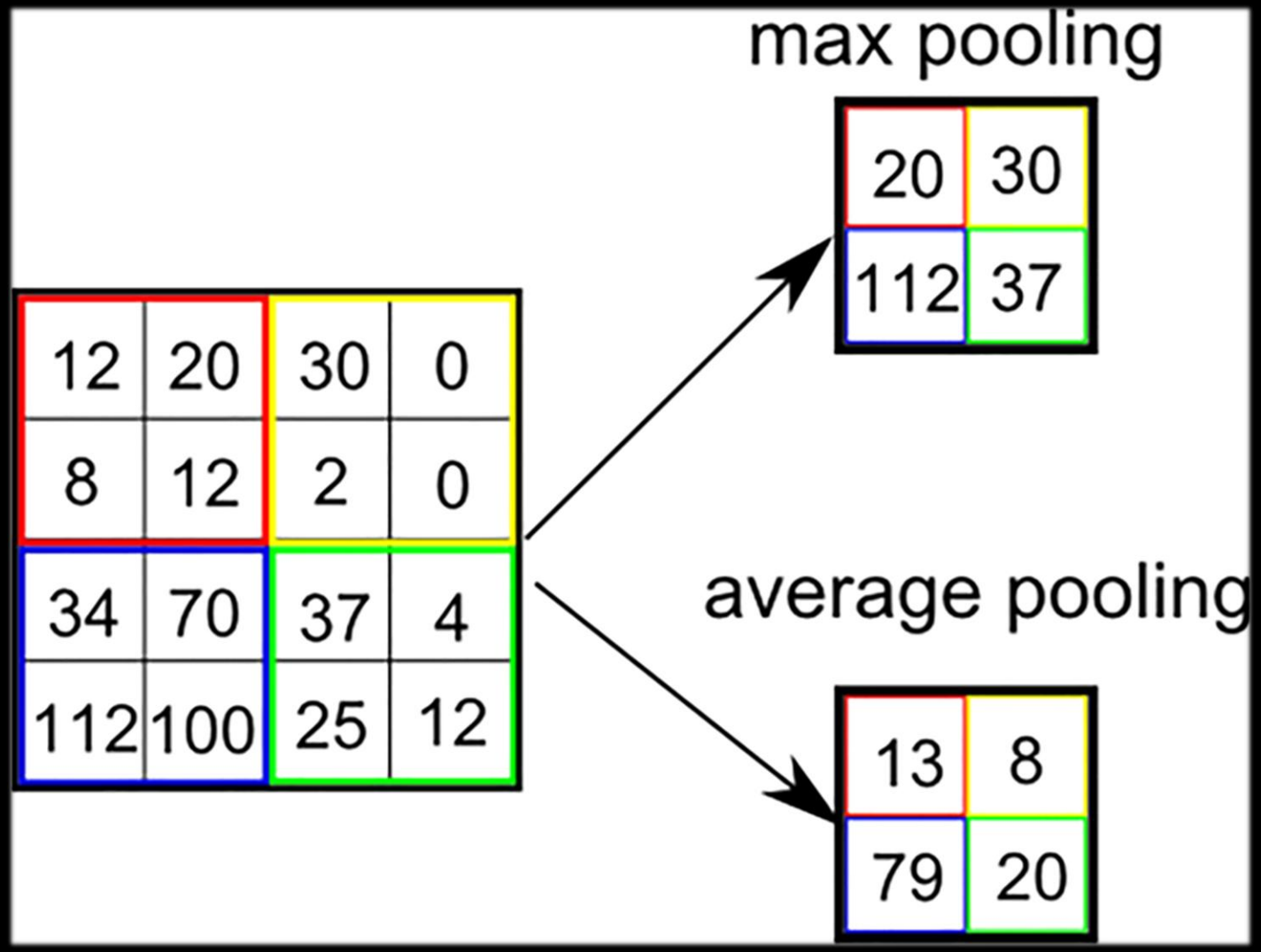
CNN Katman Yapısı

Özellik Çıkarımı: Desenleri ve şekilleri yakalar.
Sınıflandırma: Veriyi anlamlı sonuca dönüştürür.



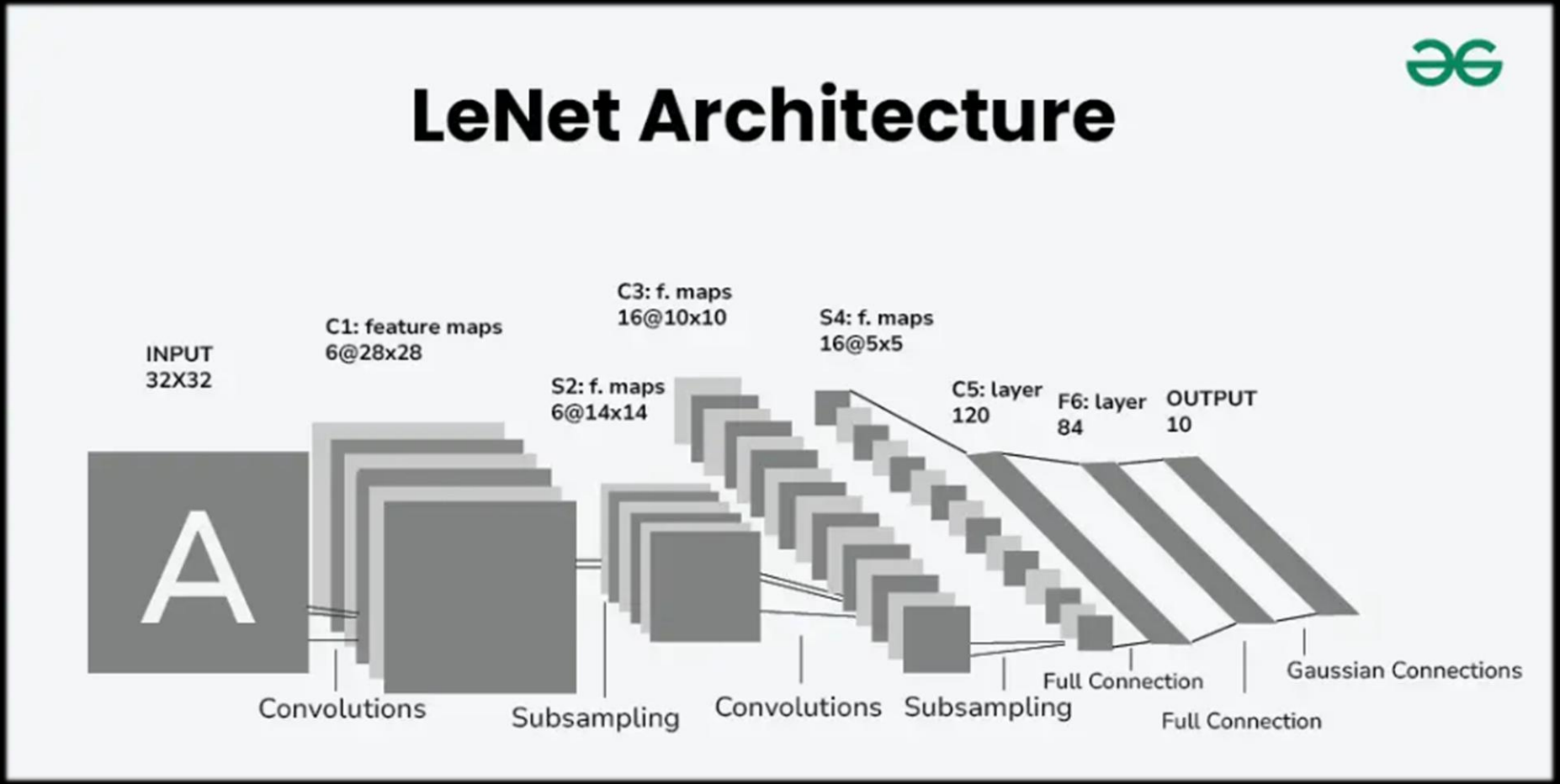
Pooling (Havuzlama)

Boyut Azaltma: Veriyi sıkıştırarak işlem hızını artırır.
Max Pooling: Bölgedeki en baskın (yüksek) özelliği seçer.
Özellik Koruma: Önemli bilgiyi kaybetmeden boyutu küçültür.



Basit CNN Mimarisi

LeNet-5: Temel ve öncü model
Katmanlı Akış: Veriyi basitten karmaşığa işler.
Tahmin Aşaması: Sınıflandırma ile sonuç üretir.



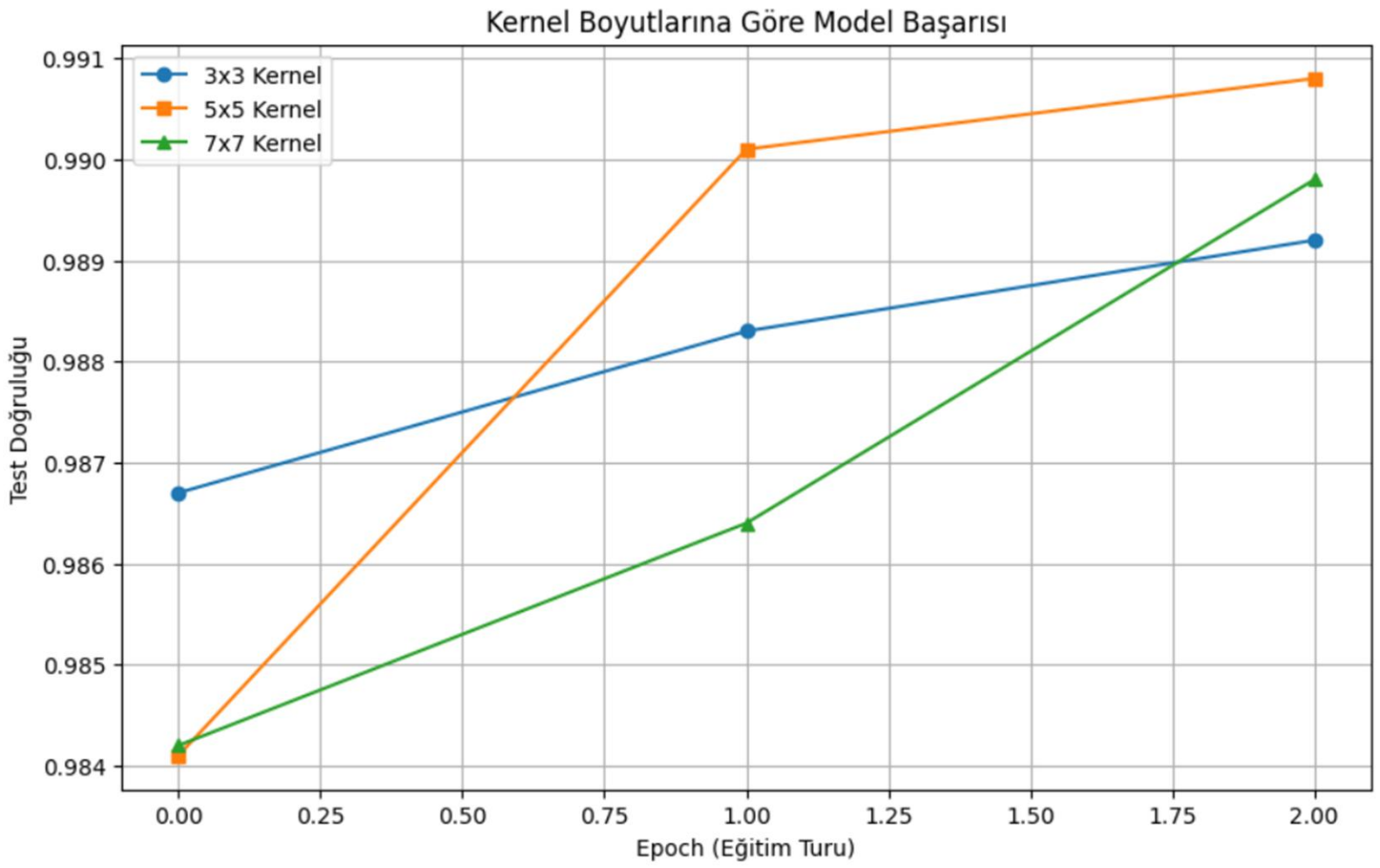
3x3, 5x5 ve 7x7 Kernel: Model Eğitim Süreci

```
3x3 Modeli Eğitiliyor...
Epoch 1/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 10s 5ms/step - accuracy: 0.9576 - loss: 0.1390 - val_accuracy: 0.9867 - val_loss: 0.0440
Epoch 2/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9853 - loss: 0.0466 - val_accuracy: 0.9883 - val_loss: 0.0344
Epoch 3/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9898 - loss: 0.0323 - val_accuracy: 0.9892 - val_loss: 0.0321

5x5 Modeli Eğitiliyor...
Epoch 1/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 10s 5ms/step - accuracy: 0.9622 - loss: 0.1248 - val_accuracy: 0.9841 - val_loss: 0.0473
Epoch 2/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9874 - loss: 0.0402 - val_accuracy: 0.9901 - val_loss: 0.0304
Epoch 3/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9912 - loss: 0.0285 - val_accuracy: 0.9908 - val_loss: 0.0271

7x7 Modeli Eğitiliyor...
Epoch 1/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 10s 5ms/step - accuracy: 0.9546 - loss: 0.1450 - val_accuracy: 0.9842 - val_loss: 0.0523
Epoch 2/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - accuracy: 0.9855 - loss: 0.0488 - val_accuracy: 0.9864 - val_loss: 0.0447
Epoch 3/3
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - accuracy: 0.9892 - loss: 0.0345 - val_accuracy: 0.9898 - val_loss: 0.0325
```

Performans Karşılaştırma Analizi




```
3x3 Toplam Parametre: 121930
5x5 Toplam Parametre: 121930
7x7 Toplam Parametre: 121930
```

Dinlediğiniz için teşekkürler.